

NLP Course Project — Research-Backed Recommendations

5-Person Team | Flagship Selection Report

How this was researched

Before ranking anything, I pulled current literature (2024–2026) and dataset availability across nine candidate problem spaces: legal contract review, cyber threat intelligence extraction, scientific citation verification, systematic-review/PICO automation, legal-document simplification, code-mixed/Indian-language grievance processing, regulatory compliance gap detection, cross-document event/claim provenance, and scam-victim forensic analysis. Four were dropped outright (see "Rejected" section) for being oversaturated, over-served by commercial tools, or too thin on data. Five survived scrutiny and are ranked below.

Ranking at a glance

Rank	Project	NLP Depth	Novelty	Data Availability	Feasibility (5 people)	Resume Value
1	Scam-Victim Forensic Schema Extractor	High	Very High	Good (public, needs assembly)	High	Very High
2	Citation-Integrity Verifier	High	High	Very Good (ready-made benchmark)	Very High	High
3	DPDP Act Compliance Gap Detector	Medium-High	Very High (timing)	Medium (build-your-own corpus)	Medium	Very High
4	Cross-Document Claim Provenance Tracker	High	High	Medium	Medium	High
5	Multi-Report CTI Campaign Aggregator	High	Medium	Good	Medium	Medium-High

My recommendation, if you want one flagship pick: **#1**. Reasoning is at the bottom of this document.

1. Scam-Victim Forensic Schema Extractor — *Recommended Flagship*

Problem & who uses it

Victim-facing fraud reports (cybercrime helplines, bank fraud desks, "romance scam" and "pig-butchering" hotlines, consumer-protection trackers) collect huge volumes of free-text narratives describing how a scam unfolded. Investigators and helpline staff currently read these manually to identify *which manipulation tactics were used* (urgency, authority impersonation, romance grooming, fake investment platforms) and *what actionable evidence is present* (crypto wallet addresses, payment rails, platform names). This is slow, inconsistent, and — critically — most systems today only capture contact metadata and dollar amounts, not the psychological/behavioral pattern of the scam, which is what actually helps investigators link cases and helps helpline agents ask the right follow-up questions in real time.

Direct users: national cybercrime helplines (India's 1930/cybercrime.gov.in is a concrete example), bank fraud-investigation teams, victim-support NGOs, and trust-and-safety teams at fintech/crypto platforms.

Why it matters

Global scam losses are now measured in the tens of billions annually (FTC alone reported **\$5.7B lost in 2024**, up 125% year-over-year), and India specifically logs a high and rising volume of cybercrime complaints per capita. Manual triage cannot scale with this volume, and generic "fraud/spam classifiers" don't capture *why* a scam worked — which is what's needed for both investigation and victim-support redesign.

Existing work

PsyScam (2025) — first benchmark grounding scam narratives in 9 cognitive/psychological-technique categories, built from 6 public scam-reporting platforms; defines PT classification, "scam completion," and "scam augmentation" tasks.

PreScam (2026) — studies how manipulation techniques *sequence* over a scam conversation rather than treating a report as a static bag of tactics.

Forensic Schema for Psychological Manipulation in Cyber Fraud (Wen et al., 2026, U. Washington Tacoma) — the closest prior art: a 4-category, 35-question forensic schema with 11 manipulation indicators + crypto-evidence fields, applied via LLM annotation to **10,994 real victim reports**, validated against two human annotators (LLM-human $\kappa = 0.69$, matching human-human $\kappa = 0.68$).

Pig-Butchering Scams exploratory study (2024) — manually coded 2,570 victim narratives from Chainabuse and Crypto Scam Tracker across contact method, relationship-building, fund-extraction technique, and psychological manipulation.

Adjacent (but generic) work: Hinglish cybercrime *classification* using HingBERT/HingRoBERTa on India's I4C hackathon data — useful as a baseline for what "shallow classification" looks like, and exactly the kind of output this project should go beyond.

The gap

The Forensic Schema paper's own finding is the opening: **manipulation-technique detection is reliable, but the actionable forensic detail supporting each finding is highly inconsistent across victim narratives, and blockchain/crypto identifiers are nearly absent** unless specifically prompted for. In other words — the extraction half of the problem is close to solved; the half nobody has built is a system that identifies *which* details are missing from a given narrative and actively asks the right follow-up question to close that gap. No public system does structured extraction *and* adaptive follow-up-question generation together.

Proposed NLP approach

Schema-guided structured extraction: fine-tune/prompt an open LLM (e.g., Llama-3-8B or Mistral-7B, instruction-tuned) to map a free-text scam narrative onto a fixed schema (contact vector, relationship-building tactic, manipulation techniques used, financial instrument, platform, evidence completeness per field).

Detail-gap detector: a lightweight classifier/rule-hybrid that flags schema fields with low "evidentiary support" (e.g., "cryptocurrency mentioned but no wallet ID extracted").

Adaptive follow-up question generator: given the gap map, generate 1–3 targeted, non-leading follow-up questions (this is the genuinely novel component — closing the gap the source paper identified) — implementable as a constrained generation module conditioned on the missing-field schema, evaluated for question relevance/specificity.

Optional stretch: cluster extracted schemas across reports to surface *scam campaigns* (same tactics/wallets reused across victims) — gives investigators a leads dashboard, not just per-report output.

Data plan

Primary: assemble a multi-source corpus from **PsyScam's public dataset** (github.com/KiteFlyKid/PsyScam), **Chainabuse** and **Crypto Scam Tracker** abuse reports (public, scrapeable per the pig-butchering study's methodology), and **BBB Scam Tracker / IC3** public narrative excerpts.

Use the Forensic Schema paper's 4-category/35-question schema as your annotation starting point (adapt/simplify to fit a semester timeline) rather than designing one from scratch.

LLM-assisted annotation with human spot-checking (as the source paper did) is a realistic, defensible annotation strategy for a student team — document inter-annotator agreement (κ) the same way the reference papers do.

If pursuing the India angle specifically, note that I4C's National Cyber Crime Reporting Portal data is *not* openly downloadable outside sanctioned hackathons — treat this as a stretch/context source, not your primary corpus.

Evaluation plan

Schema-field extraction: per-field F1/accuracy against held-out human-annotated narratives; report macro-F1 across the 11 manipulation indicators.

Agreement: LLM-vs-human κ , benchmarked against the 0.69 figure from the Forensic Schema paper as your baseline to beat or match.

Follow-up question quality: human evaluation (relevance, non-leading-ness, whether it actually elicits the missing field) — this is your headline novel metric since no baseline exists for it.

Downstream utility: simulate investigator triage time reduction (manual field-completion time vs. system-assisted).

Expected prototype output

A web tool where a victim/agent pastes or dictates a scam narrative and receives: a structured forensic case file (schema fields populated with confidence + evidence spans), a flagged list of missing/weak evidence fields, and 1–3 generated follow-up questions to ask next. A secondary "campaign view" clusters similar cases.

Novelty vs. generic projects

This is explicitly **not** spam/fraud binary detection (which the prompt asked to avoid) — the output is a structured forensic case file plus an interactive gap-closing mechanism, grounded in a specific, named limitation from a 2026 paper. Very few student projects will have found and be building on a gap this recent.

Difficulty & feasibility for a 5-person team

Realistic. Split naturally: (1) data collection/scraping + schema adaptation, (2) extraction model fine-tuning, (3) gap-detection + question-generation module, (4) evaluation/annotation coordination, (5) frontend + campaign-clustering view. No need to train models from scratch — pretrained LLMs + fine-tuning/LoRA is sufficient and matches the reference papers' own methodology.

Resume/interview value

Very high. "We read the newest fraud-forensics literature, found a documented gap (inconsistent evidentiary detail in victim reports), and built the missing half of the pipeline — an adaptive follow-up system — evaluated against real annotator-agreement baselines" is a strong, specific, defensible story, and ties naturally to India's cybercrime context if you want that framing.

Key references

Wen et al., *Forensic Schema for Psychological Manipulation in Cyber Fraud: LLM-Driven Victim Reports Analysis*, arXiv:2607.07751 (2026)

PsyScam: A Benchmark for Psychological Techniques in Real-World Scams, arXiv:2505.15017 (2025) — github.com/KiteFlyKid/PsyScam

PreScam: A Benchmark for Predicting Scam Progression from Early Conversations, arXiv:2605.12243 (2026)

An Explorative Study of Pig Butchering Scams, arXiv:2412.15423 (2024)

Rani et al., *Automated Classification of Cybercrime Complaints using Transformer-based Language Models for Hinglish Texts*, arXiv:2412.16614 (2024) — baseline reference for "shallow classification"

2. Citation-Integrity Verifier

Problem & who uses it

When a paper cites a source to support a claim, the citation is often bibliographically valid but **semantically misaligned or outright wrong** — the cited work doesn't actually say what's claimed. This is distinct from "does this claim match reality" (fact-checking); it's "does this citation's target actually support this specific sentence." Users: journal editors/peer reviewers, PhD students and labs doing literature reviews, publishers building integrity-screening tools, and — increasingly — anyone reviewing AI-assisted writing, where LLM-generated citations are a known failure mode.

Why it matters

A 2024 biomedical corpus study found **39.18% of annotated citation instances contained accuracy errors** — nearly 2 in 5. As LLM-assisted paper writing grows, unchecked citation drift is a rising integrity risk, not a shrinking one.

Existing work

Citation-Integrity corpus (Oxford Bioinformatics, 2024): 100 highly-cited biomedical papers, 3,063

annotated citation instances (ACCURATE/NOT_ACCURATE/IRRELEVANT), public at github.com/ScienceNLP-Lab/Citation-Integrity.

SciFact / SciFact-Open (Wadden et al., 2020/2022): the foundational scientific claim-verification benchmark (1.4K claims vs. abstracts, later expanded to 500K-abstract retrieval).

DeepSciVerify (2026): LLM-driven evidence-escalation pipeline for claim-citation alignment, evaluated on the SCitance dataset.

SemanticCite (2025): 1,000+ citations across 8 disciplines with functional/semantic annotations.

CiteME (NeurIPS 2024 Datasets & Benchmarks): shows even strong LMs achieve only ~35% accuracy identifying the correctly cited paper from an excerpt.

The gap

The Citation-Integrity paper's own best system — a retriever + fine-tuned claim-verification model — reaches only **0.59 micro-F1 / 0.52 macro-F1**; GPT-4 in-context learning does *better* on accurate citations but *worse* on catching errors (0.65 micro-F1, but only 0.45 macro-F1, meaning it under-catches the erroneous class that matters most). There is no deployed, usable tool that flags misaligned citations while a paper is being written or reviewed — only research-stage classifiers evaluated on held-out corpora.

Proposed NLP approach

Retrieval stage: given a citing sentence + its reference, retrieve the top-k evidence sentences from the cited paper (BM25 + dense re-ranker, following the Citation-Integrity paper's own best pipeline as your baseline).

Verification stage: fine-tune a NLI/claim-verification model (e.g., a SciFact-pretrained MultiVerS-style model or a smaller open LLM) to classify ACCURATE / NOT_ACCURATE / IRRELEVANT, with rationale-sentence highlighting for explainability.

Differentiator: combine the biomedical corpus (which is narrow) with a CS/NLP-domain extension you build yourselves (since ACL/arXiv papers are freely available and self-citation-checkable), demonstrating cross-domain generalization — a genuine open question in the literature.

Optional: severity scoring (minor paraphrase drift vs. outright unsupported claim) rather than a flat 3-way label, which none of the cited systems currently do.

Data plan

Primary: the public Citation-Integrity corpus (ready-made, expert-annotated, no collection needed).

Secondary/extension: build a small CS-domain test set from ACL Anthology + arXiv papers (self-supervising via citation sentence vs. cited abstract pairs, following SciFact's citance-based annotation protocol) to test cross-domain robustness — this "we built our own eval slice" step is what differentiates you from simply re-running the existing benchmark.

Evaluation plan

Micro/macro-F1 against the Citation-Integrity test split, directly comparable to the paper's published 0.59/0.52 baseline and GPT-4 ICL's 0.65/0.45.

Retrieval quality: Recall@k / MRR for the evidence-retrieval stage.

Cross-domain generalization: same metrics on your self-built CS/NLP slice.

Human evaluation of rationale quality (does the highlighted evidence span actually justify the verdict?).

Expected prototype output

A tool that takes a paper (or a citing sentence + BibTeX/DOI) and returns a per-citation accuracy verdict with a highlighted evidence span from the cited work and a confidence/severity score — usable as a Word/LaTeX plugin concept or a standalone web checker.

Novelty vs. generic projects

Not a "PDF summarizer" or generic fact-checker — it's a narrowly scoped, benchmarked task (claim–citation alignment) with a *specific, quantified, still-open performance gap* you can cite and try to beat.

Difficulty & feasibility

High feasibility — dataset is ready-made, task is well-defined, and the retrieval+NLI architecture is standard and well-documented in the cited papers. Very achievable polish level for a semester.

Resume/interview value

Strong and cleanly explainable: "we benchmarked against a published 0.59 F1 result and built a cross-domain extension" is a crisp, verifiable claim for interviews.

Key references

Assessing citation integrity in biomedical publications: corpus annotation and NLP models, Bioinformatics 40(7), 2024 — github.com/ScienceNLP-Lab/Citation-Integrity

Wadden et al., *Fact or Fiction: Verifying Scientific Claims*, EMNLP 2020 (SciFact)

DeepSciVerify: Verifying Scientific Claim–Citation Alignment via LLM-Driven Evidence Escalation, arXiv:2605.27710 (2026)

SemanticCite: Citation Verification with AI-Powered Full-Text Analysis, arXiv:2511.16198 (2025)

CiteME: Can Language Models Accurately Cite Scientific Claims?, NeurIPS D&B 2024

3. DPDP Act Compliance Gap Detector (India-specific)

Problem & who uses it

India's **Digital Personal Data Protection Rules, 2025** were notified on 13 November 2025, starting an 18-month phased compliance countdown for every Indian business handling personal data. Companies must now check whether their internal privacy policies, consent flows, and vendor contracts actually satisfy the Act's obligations — today this is done manually by compliance consultants (KPMG, Deloitte, etc. are actively publishing manual "action plan" guides). Users: startup/SME compliance teams, DPOs, legal-tech vendors, and consulting firms without dedicated NLP tooling yet.

Why it matters

This is a live, dated regulatory event with real financial stakes (penalties up to ₹250 crore per breach) and a hard compliance clock — a rare case of a course project mapping to something genuinely time-sensitive rather than evergreen.

Existing work

OPP-115 (Wilson et al., 2016), **PolicyQA** (2020), **PolicyIE** (2021), **PrivacyGLUE** (2023): established privacy-policy NLP corpora and tasks — but built for *US* privacy policy conventions (CCPA-era), not India's Act.

NLP-based Automated Compliance Checking of Data Processing Agreements against GDPR (2022): the closest transferable methodology — sentence-level semantic matching of DPA text against 45 GDPR requirements, evaluated on 54 real DPAs.

ComplianceNLP (2026) and **RAGulating Compliance** (2025): recent KG-augmented RAG systems for multi-framework regulatory gap detection (SEC/MiFID/Basel) — architecture template you can adapt.

One very early **arXiv preprint (Jan 2026)** proposes an agentic framework for DPDP-aligned data governance — confirms the space is *just* opening up, with no established benchmark or public dataset yet.

The gap

There is currently **no public, benchmarked obligation-extraction or gap-detection dataset for the DPDP Act** — everything published on GDPR/CCPA doesn't transfer cleanly because DPDP has India-specific constructs (Data Fiduciary/Data Principal terminology, Consent Managers, the Data Protection Board, distinct breach-notification and cross-border-transfer rules). This is a textbook

"research-to-product gap": methodology exists (the GDPR DPA-checking paper), but the domain-specific resource doesn't.

Proposed NLP approach

Obligation extraction: parse the DPDP Act + Rules text into a structured obligation ontology (subject/actor, obligation type, condition, deadline) — NER + relation extraction, following the GDPR DPA-checker's sentence-to-requirement matching design.

Policy-to-obligation matching: given a company's privacy policy / internal data-handling doc, semantically match each policy statement to the obligation(s) it satisfies (sentence embeddings + entailment/NLI scoring), following the 2022 GDPR paper's protocol directly.

Gap report generation: for every unmatched or weakly-matched obligation, generate a structured gap report with severity (e.g., missing breach-notification clause = high severity; missing optional disclosure = low).

This is the one project on this list where **you must build your own annotated dataset** — treat that as a deliverable, not an obstacle: annotate ~30–50 real/sample privacy policies (many Indian companies have published DPDP-readiness policy updates you can use as source text) against the Act's clauses, following the 2022 GDPR paper's two-phase manual-then-automated annotation protocol as your template.

Data plan

Primary source text: the DPDP Act, 2023 + DPDP Rules, 2025 (public, MeitY).

Policy corpus: publicly available Indian company privacy policies + consulting firms' published "DPDP readiness" checklists (KPMG, Deloitte have public guidance documents breaking the Act into implementation-ready obligation categories — useful as a semi-structured starting ontology, not to be copied verbatim).

Annotation: your team defines and labels obligation-to-policy-clause mappings; budget meaningful time for this (2022 GDPR paper's team spent two working weeks on it for 24 documents), and document inter-annotator agreement.

Evaluation plan

Obligation-extraction F1 against your hand-labeled test set.

Policy-matching precision/recall (does the system correctly flag when a policy clause does/doesn't satisfy an obligation?), compared against a keyword-matching baseline.

Gap-report usefulness: human evaluation by comparing your system's gap report against a manually produced one for a held-out company policy.

Expected prototype output

Upload a privacy policy → receive a structured DPDP compliance gap report: obligation-by-

obligation coverage status, missing-clause flags with severity, and suggested clause language pulled from the Act's own text.

Novelty vs. generic projects

Not a generic "RAG chatbot over a legal PDF" — the deliverable is a structured, severity-scored gap report against a specific, freshly-enacted regulation with no existing benchmark, which is a defensible and rare claim.

Difficulty & feasibility for a 5-person team

Medium — the main cost is annotation time (you're building the first dataset of its kind), which is real work but scoped and bounded; the modeling techniques themselves (NER, sentence-matching, NLI) are standard. This is the idea most dependent on your team being willing to do careful manual annotation early.

Resume/interview value

Very high, especially for an India-based audience or any legal-tech/compliance-adjacent interview: "we built the first obligation-extraction dataset and gap-detection system for India's new data protection law, during its live 18-month compliance window" is distinctive and timely.

Key references

Wilson et al., *OPP-115 Corpus*, 2016; Ahmad et al., *PolicyQA*, 2020 / *PolicyIE*, 2021

NLP-based Automated Compliance Checking of Data Processing Agreements against GDPR, arXiv:2209.09722 (2022)

ComplianceNLP: Knowledge-Graph-Augmented RAG for Multi-Framework Regulatory Gap Detection, arXiv:2604.23585 (2026)

RAGulating Compliance: A Multi-Agent Knowledge Graph for Regulatory QA, arXiv:2508.09893 (2025)

An Agentic Software Framework for Data Governance under DPDP, arXiv:2601.01101 (2026)

Digital Personal Data Protection Rules, 2025 (MeitY, notified 13 Nov 2025)

4. Cross-Document Claim & Narrative Provenance Tracker

Problem & who uses it

When a claim spreads across news outlets, social media, and blogs, it mutates — details get added, dropped, or distorted at each retelling. Journalists and researchers currently have no structured way to trace *where a claim originated* and *how it changed* as it propagated across sources; they either read everything manually or rely on binary true/false fact-checking, which discards the provenance information entirely. Users: investigative journalists, misinformation researchers, fact-checking organizations, OSINT analysts.

Why it matters

This is explicitly framed as something *other than* fake-news classification — the output is a structured attribution graph and timeline, which is more useful for investigation than a single veracity label, and is closer to how journalists actually work (tracing a claim back to its first appearance and identifying where it was distorted).

Existing work

CRAB (2024): cross-document dataset (173 documents, 2,730 event pairs, 20 stories) explicitly built to assess causal-relationship strength between events *across* different articles, not just within one.

CDEC — Cross-Document Event Coreference (2021): 21K Wikinews articles clustered into storylines via a "Related News" hyperlink graph, with dense cross-document event-identity annotation.

CLES (2024): large-scale cross-document event extraction dataset (20K+ documents, 37.7K events, 70%+ cross-document) — built for Chinese, which is a genuine limitation/opportunity (an English-language equivalent doesn't yet exist at this scale).

TIMELINE (2023) and **Background Summarization of Event Timelines** (2023, using Timeline17/Crisis/Social Timeline datasets): temporal-ordering and timeline-construction resources.

The gap

Every one of the datasets above is framed as "cross-document event/temporal extraction is still a fundamentally under-explored area" relative to single-document extraction — and none of them combine event linking with *claim-level* attribution (who said it first, how the wording changed) in

English at scale. The CLES dataset explicitly calls this an "unexplored" paradigm and is Chinese-only, leaving an open English-language opportunity.

Proposed NLP approach

Document clustering: group articles/posts discussing the same underlying story (embedding-based clustering + the "related-article graph" trick from CDEC).

Cross-document event/claim coreference: identify when two documents are describing the same underlying claim (fine-tune a cross-encoder on CDEC/CRAB-style pairs).

Temporal ordering + provenance graph construction: order claim mentions by publication date, and represent claim mutation as a graph (nodes = claim instances per source, edges = "derived from" with a distortion/similarity score).

Distortion scoring: quantify how much a claim instance diverges from its earliest known instance (semantic similarity + added/dropped-entity detection) — this is your novel contribution layered on top of existing cross-doc coreference work.

Data plan

CDEC (Wikinews, public, CC-BY) and CRAB are directly usable for training/evaluating the coreference and causal-strength components.

For the claim-provenance/timeline layer, construct your own small corpus around 5–10 real news stories (scrape via RSS/NewsAPI-style public feeds, respecting terms of service) and manually annotate claim mutation — a bounded, achievable task for one sub-team.

Evaluation plan

Cross-document coreference: precision/recall against CDEC's gold clusters.

Causal/temporal ordering accuracy against CRAB and TIMELINE.

Provenance-graph quality: human evaluation of whether the constructed "who said it first, how did it change" graph matches a manually researched ground truth for your own annotated stories.

Expected prototype output

Given a claim or topic, the system outputs a timeline + attribution graph showing the earliest known source, subsequent restatements, and where/how the wording or details diverged — visualized as an interactive graph, not a single true/false verdict.

Novelty vs. generic projects

Explicitly differentiated from fake-news detection by design: no veracity label is produced at all — only structured provenance, which is a genuinely underused framing per the literature itself.

Difficulty & feasibility for a 5-person team

Medium — the individual components (coreference, temporal ordering) are backed by existing datasets, but stitching them into a coherent end-to-end provenance graph and building your own annotated evaluation slice is real integration work. Doable, but the most engineering-heavy of the five ideas.

Resume/interview value

High, particularly for anyone interested in journalism-tech, OSINT, or misinformation research — a distinctive graph-output framing that avoids the "we built another fake-news classifier" cliché.

Key references

CRAB: Assessing the Strength of Causal Relationships Between Real-world Events, arXiv:2311.04284 (2024)

Cross-document Event Identity via Dense Annotation, arXiv:2109.06417 (2021) — CDEC

Harvesting Events from Multiple Sources: Towards a Cross-Document Event Extraction Paradigm, arXiv:2406.16021 (2024) — CLES

TIMELINE: Exhaustive Annotation of Temporal Relations..., arXiv:2310.17802 (2023)

Background Summarization of Event Timelines, arXiv:2310.16197 (2023)

5. Multi-Report CTI Campaign Aggregator

Problem & who uses it

Security analysts read cyber threat intelligence (CTI) reports and manually tag which MITRE ATT&CK tactics/techniques a threat actor used, so defenders can map detections to real adversary behavior. Users: SOC analysts, threat-intel teams, MSSPs.

Why it matters

This directly supports proactive defense — accurately mapped CTI is what lets defenders build detections ahead of an attack rather than after one.

Existing work

This is the most heavily researched space on this list: **TRAM** (MITRE's own dataset), **CTIBench** (NeurIPS 2024), **CTI-HAL** (2025), **CTINEXUS**, **CyBERTa**, and dozens of BERT/LLM-based single-

report TTP extractors. A **2025 USENIX Security "SoK" paper explicitly asks "Automated TTP Extraction from CTI Reports: Are We There Yet?"** — surveying the entire field.

The gap

The SoK paper's answer is effectively "not yet," and the specific, still-open sub-problem is **cross-report, campaign-level aggregation**: almost all existing work tags techniques *within a single report*, but real adversary campaigns are described piecemeal across dozens of reports from different vendors, with inconsistent naming and partial overlap. A 2026 paper, *Beyond Single Reports: Evaluating Automated ATT&CK Technique Extraction in Multi-Report Campaign Settings*, is one of the only works to test this directly, and shows performance degrades substantially versus single-report extraction.

Proposed NLP approach

Single-report technique extraction: fine-tune a transformer (e.g., SecureBERT/CyBERTa-style) on TRAM/CTI-HAL for per-report ATT&CK tagging — this is your baseline, well-supported by existing code/models.

Cross-report entity/campaign linking: cluster reports describing the same campaign or actor (via shared IOCs, actor-name aliasing, temporal proximity) using entity resolution techniques.

Aggregated technique-profile construction: merge per-report technique tags into a single deduplicated, confidence-weighted campaign profile, resolving naming inconsistencies across vendors (a known, cited pain point).

Output as a STIX-compatible structured campaign summary.

Data plan

TRAM (MITRE Center for Threat-Informed Defense, public GitHub), CTIBench, CTI-HAL — all public, ready to use for the single-report baseline.

For the cross-report aggregation layer, construct a small multi-report campaign test set (public APT threat reports referencing the same named campaign from multiple vendors — APTnotes repository is a public aggregator) and manually verify aggregated technique coverage.

Evaluation plan

Single-report baseline: F1 against TRAM/CTIBench published numbers.

Cross-report aggregation: precision/recall of the merged campaign profile against manually verified ground truth, plus a "how much does performance degrade going single→multi-report" analysis, directly comparable to the 2026 "Beyond Single Reports" paper's findings.

Expected prototype output

Given a set of CTI report URLs/PDFs about the same campaign, output one unified, deduplicated ATT&CK technique profile with confidence scores and source attribution per technique.

Novelty vs. generic projects

Positioned specifically at the cross-report aggregation gap rather than re-solving already well-covered single-report tagging — but this is the most competitive/crowded space on the list, so novelty depends on staying disciplined about that scoping.

Difficulty & feasibility

Medium — strong existing tooling lowers the floor (achievable baseline is fast to stand up), but the genuinely novel cross-report layer requires real entity-resolution engineering.

Resume/interview value

Medium-high — cybersecurity-adjacent NLP is a strong recruiting signal, but because the space is crowded, be ready to clearly articulate why your project isn't "yet another ATT&CK tagger."

Key references

Büchel et al., *SoK: Automated TTP Extraction from CTI Reports — Are We There Yet?*, USENIX Security 2025

Beyond Single Reports: Evaluating Automated ATT&CK Technique Extraction in Multi-Report Campaign Settings, arXiv:2604.07470 (2026)

CTI-HAL: A Human-Annotated Dataset for Cyber Threat Intelligence Analysis, arXiv:2504.05866 (2025)

MITRE Center for Threat-Informed Defense, *TRAM dataset*, github.com/center-for-threat-informed-defense/tram

Alam et al., *CTIBench: A Benchmark for Evaluating LLMs in Cyber Threat Intelligence*, NeurIPS 2024

Considered and rejected

Per your instruction to be critical rather than padding the list:

Legal contract clause review/retrieval (CUAD/ACORD/MAUD) — excellent datasets and research depth, but this space is now heavily commercialized (Kira Systems, LawGeex, Ironclad, Luminance, etc.) and extremely well-covered academically. A student prototype has little room to

say anything a dozen funded startups and multiple NeurIPS-track datasets haven't already said. Reject for novelty and differentiation, not for data or feasibility.

PICO extraction / systematic-review automation — strong datasets (EBM-NLP) but the practical tooling gap is mostly closed: Covidence, DistillerSR, RobotReviewer and multiple GenAI pipelines (one 2024 study reported 98% GPT-4o extraction accuracy on 680K+ PICOs) already serve this well. Reject as over-served.

Legal/bureaucratic plain-language simplification — real research gap exists (small, thin datasets like the Plain-English-Contracts corpus), but the deliverable shape is uncomfortably close to "generic PDF summarizer," which you explicitly asked to avoid, and the evaluation story (readability + faithfulness) is harder to make as crisp as the other five. Reject as too close to the "slop" category and too thin on data.

Plain Hinglish/code-mixed cybercrime complaint classification — good data story (I4C hackathon dataset, HingBERT), but as a bare classification task it's exactly the "basic text classification with an accuracy score" pattern you asked to avoid, and the I4C dataset itself isn't openly accessible outside sanctioned hackathons. The *idea* survives in a stronger, more actionable form as Project #1 above (structured extraction + follow-up generation on public scam-report data), which I'd treat as the better version of this same instinct.

My actual recommendation

If you want one flagship: **Project #1, the Scam-Victim Forensic Schema Extractor.**

Reasoning, briefly: it's the only idea on this list built on genuinely fresh (2025–2026) research with a *named, specific, still-open limitation* you can directly target (the "forensic detail gap" from the Wen et al. paper) rather than a broad, long-standing open problem where you're one of a hundred groups chipping away at it. The data story is realistic without needing restricted government access. The output — a structured case file plus an adaptive follow-up-question generator — is unambiguously "actionable," not a bare classification score, which satisfies your hardest constraint. And the interview story writes itself: you can point to a specific paper, a specific gap sentence in that paper, and a specific system you built to close it.

Project #2 (Citation Integrity) is the safest, highest-certainty-of-success alternative if your team wants a lower-risk build with a ready-made benchmark and less annotation overhead — recommend it as backup if #1's data-assembly work feels too open-ended for your timeline.

Project #3 (DPDP) is the highest-ceiling choice if your team is comfortable doing real annotation work and wants maximum resume distinctiveness for the Indian job market specifically — but it's the most annotation-heavy of the five, so only choose it if you're confident in your team's discipline around that work.